Jodh Singh Nahal

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Education

University of British Columbia *Vancouver, BC*
Bachelor of Applied Science in Computer Engineering *Expected May 2028*
- Presidential Scholars Award ($60,000), Dean's Honour List, available 4 or 8 months, CGPA: 84.7%.

Technical Skills

Languages: C/C++, Python, Java, Lua, JavaScript
Graphics & Simulation: Bresenham Rendering, VGA Graphics, MuJoCo, dm_control, PyGame, OpenCV
Tools: Git, Linux, GDB, Docker

Experience

Hardware Verification Engineering Intern *Vancouver, BC*
AMD *Sept 2026 – Present*
- Debugging RTL behavior through simulation and waveform analysis.
- Contributing to verification of the IOHUB subsystem for next-generation AMD processor architectures.
- Verifying IOMMU L1 cache behavior using UVM in a constrained-random testbench.
- Expanded the constrained-random testbench to cover corner cases and improve functional coverage.

Software Test / Embedded Development Co-op *Burnaby, BC*
Xantrex *May 2024 – Aug 2025*
- Achieved 7× device capacity, 90% lower CPU usage, and ∼2 s lower latency after the C++ migration.
- Developed an object-oriented, multi-threaded, non-blocking C++ firmware architecture on Linux-based Raspberry Pi gateways to parse RV-C CAN frames from STM32 devices and publish structured messages over MQTT.
- Replaced a ∼60k-line Python implementation with the C++ architecture.

Firmware / Embedded Systems Developer *Vancouver, BC*
UBC Rover *Sept 2023 – July 2026*
- Built a ROS–Arduino pipeline using rosserial, OpenCV, and Python nodes to modulate USB camera framerates from potentiometer readings.
- Automated firmware validation with GitHub Actions CI and Catch2 unit tests to improve regression coverage.
- Led a team of 3 engineers building a distributed embedded control system across ESP32 (FreeRTOS), STM32, Arduino, and TI LaunchPad MCUs over SPI/I2C.
- Aggregated inputs from switches, buttons, potentiometers, encoders, and joysticks and streamed data over UART.
- Developed a 16-channel telemetry system on a TI C2000 (LAUNCHXL-F28069M) using ePWM-triggered ADC interrupts, streaming GPIO/ADC data over SCI/UART.

Undergraduate Research Assistant *Vancouver, BC*
UBC Electrical and Computer Engineering — Supervisor: Dr. Lutz Lampe *May 2026 – Sept 2026*
- Optimizing power extraction of piezoelectric energy harvesting interface circuits (SEH, SSHI, SECE) through LTspice and Simulink/Simscape simulation.
- Researching battery-less platforms for the Internet of Underwater Things via simulation of piezoelectric interface circuits.
- Designing embedded power-management PCB hardware in KiCad integrating AC–DC rectifiers, transformers, and switching circuitry.

Projects

Graphics Rendering & VGA Display | *SystemVerilog, FPGA, VGA Adapter*
- Implemented graphics rendering pipelines and Bresenham circle-drawing in SystemVerilog for a 160×120 VGA display on a DE1-SoC FPGA.
- Designed RTL datapaths for single-cycle-per-pixel rendering and debugged timing and logic issues via waveform inspection.

Image Compression Quadtree (Toqutree) | *C++, STL, Image Processing*
- Engineered a modified quadtree in C++ that recursively decomposes images for color quantization and compression.